

Bayesian analysis of field goal attempts in American football

Gavin Murray

BE 4910

Dr. Rego

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Introduction:

I chose to study field goal accuracy in the NFL because it's a critical part of the game that often determines the outcome, especially in close matchups. Coaches are often faced with the high-stakes decision to attempt a field goal or go for it on 4th down in order to win a game. Teams such as the Lions and Commanders have embraced aggressive strategies, opting to risk going for it on 4th down each time pursuit of scoring a touchdown. Meanwhile, others like the Steelers in the 2024 season have leaned heavily on their kickers, often securing wins by relying almost exclusively on field goals. With kickers now consistently making longer attempts, understanding the factors behind successful field goals have never been more relevant.

As a fan of football, I was curious how field goal success depends on distance and at what distances teams decided to attempt kicks. Initially I wanted to compare how field goal success rates and attempt distributions changed over time by comparing the early 2000s (2002-2004) and modern NFL era (2022-2024). This was to analyze how kicker's accuracy evolved over time and how teams attempted longer kicks over the years. Ultimately, I decided that choosing to study just one year (2024) would be a more effective approach as it allows me to assume independence of kicks, consistent bin distributions, and stable logistic regression parameters. Although, I did not pursue other prior studies while doing research of this project, most data resources did not consider an underlying distribution of attempt distances or a joint model of attempts and success probabilities using a Bayesian framework. This project was implemented in order to fill the gap of modeling attempt distribution and success probability using Bayesian inference. This allows for a more complete understanding of decision-making under uncertainty in football strategy.

Methods

To analyze NFL field goal accuracy, I collected 2024 season field goal data from the NFL database. From this dataset, I extracted each field goal attempt's distance and whether or not it was made or missed. I aggregated this data into 5 distance bins (0-19.5, 19.5-29.5, 29.5-39.5, 39.5-49.5, and 50-100 yards) to create counts of attempts n_k and made field goals m_k in each bin k allowing for grouped statistical modeling. This implemented the goal of modeling the two components: The probability of a field goal being made as a function of distance and the distribution of field goal attempt distances across the field. To achieve this, I used a Bayesian joint model that captures both success probabilities and the attempt distribution.

First, to model the data, let $X \in (0, 1)$ denote the normalized field goal distance that is equal the actual field goal distance divided by 100 yard. Let X follow a beta distribution with parameters α and β .

$$X \sim \text{Beta}(\alpha, \beta).$$

The data provided by the NFL database does not provide an actual X value for each kick but rather how many X values fall within each of the 5 bins. From this a proportion of X values that fall within a bin k , ϕ_k , can be found by integrating the Beta density over that bin. This converts the continuous Beta density into a discrete probability vector over the bins. Let a_k denote the lower-bound distance for bin k and b_k denote the upper-bound distance for bin k .

$$\phi_k = \int_{a_k}^{b_k} f_{\text{Beta}}(x|\alpha, \beta) dx = F_{\text{Beta}}(b_k|\alpha, \beta) - F_{\text{Beta}}(a_k|\alpha, \beta)$$

From this equation, we can find ϕ_k by taking the difference between the cdfs of the edge of a higher bin and a lower bin. Simplified this becomes:

$$\phi_k = F_{\text{Beta}}(b_k|\alpha, \beta) - F_{\text{Beta}}(a_k|\alpha, \beta)$$

where F_{Beta} is the Beta CDF. These bin probabilities ϕ_k form the parameters of a multinomial likelihood for attempt counts n_k . In order to model attempt counts, let n_k denote the number of field goal attempts observed in a bin k . Assuming each attempt is independently drawn from the Beta-derived probability vector $\phi = (\phi_1, \dots, \phi_K)$, then the number of attempts in each bin follows a multinomial distribution based on proportions ϕ_k :

$$n = (n_1, \dots, n_K) \sim \text{Multinomial}(N, \phi)$$

Where N = total number of attempts

In order to model successes, let m_k denote the number of field goals made observed in bin k . Conditional on n_k , m_k follows a binomial distribution where p_k is the success probability for bin k :

$$m_k | n_k \sim \text{Binomial}(n_k, p_k)$$

Then assume the probability of making a field goal depends on distance through a logistic function with parameters γ_0 and γ_1 :

$$p(x|\gamma_0, \gamma_1) = \frac{1}{1 + e^{-(\gamma_0 + \gamma_1 x)}}$$

The logistic function is used because we're modeling the probability of a success field goal as a smooth function of distance X . We assume field goal success gets harder with distance, but not linearly, rather a more sigmoidal relationship, which is captured better with a logistic function as it success probability is near 1 at short distances, dropping off quickly around a transition zone, and then approaches 0 at longer distances. The success probability for bin k is the weighted

average of $p(x|\gamma_0, \gamma_1)$ values over the interval for bin k , where the weight function is the probability density of field goals attempted

$$p_k = \frac{1}{\phi_k} \int_{a_k}^{b_k} p(x|\gamma_0, \gamma_1) f_{\text{Beta}}(x|\alpha, \beta) dx$$

Because the logistic model is continuous, the exact distances for each kick is unknown, only the fact that it lies within a distance bin is known. Average success probability within each bin is calculated through $p_k \cdot f_{\text{Beta}}(x|\alpha, \beta)$ models the distribution of attempt distances within the whole field, ϕ_k is used as the normalization constant to fit the integral to the bin, and the integrand multiplies the success probability $p(x)$ by the probability density of a kick being at that distance. This accounts for the uncertainty in exact distance within each bin, while still using the continuous model for success probability. p_k values are unknown because γ_0 and γ_1 are unknown, just like the ϕ_k values are unknown because α and β are unknown. Therefore, we must perform inference for the 4 parameters γ_0 , γ_1 , α , and β .

γ_0 and γ_1 are parameters of the logistic function modeling field goal success probability as a function of distance, while α and β are the shape parameters of the Beta distribution modeling the distribution of attempt distances across the field. In order to infer these parameters, Bayesian inference via Laplace approximation must occur.

posterior $\propto$ likelihood $\times$ prior

$$f_{\gamma_0, \gamma_1, \alpha, \beta|m, n}(\gamma_0, \gamma_1, \alpha, \beta|m, n) \propto P(m, n|\gamma_0, \gamma_1, \alpha, \beta) f_{\gamma_0, \gamma_1, \alpha, \beta}(\gamma_0, \gamma_1, \alpha, \beta)$$

From the definition of conditional probability the likelihood can further be decomposed as:

$$P(m, n|\gamma_0, \gamma_1, \alpha, \beta) = P(m|n, \gamma_0, \gamma_1, \alpha, \beta) P(n|\gamma_0, \gamma_1, \alpha, \beta)$$

Assuming all field goals are independent, $P(m|n, \gamma_0, \gamma_1, \alpha, \beta)$ is a product of the binomial PMF values, which hosts the parameters of n_k and p_k which do not depend on α or β so they can be ignored in following equation:

$$P(m|n, \gamma_0, \gamma_1, \alpha, \beta) = P(m|n, \gamma_0, \gamma_1) = \prod_{k=1}^K P_{\text{Bin}}(m_k|n_k, p_k)$$

Where P_{Bin} denotes a binomial PMF. $P(m|n, \gamma_0, \gamma_1, \alpha, \beta)$ can be read as the probability of the successes conditioned on those attempt counts and the success probabilities per bin.

Furthermore, $P(n|\gamma_0, \gamma_1, \alpha, \beta)$ is a multinomial PMF value in which the vector $n = [n_1, n_2, n_3, n_4, n_5]$ is treated as a single multinomial observation. The multinomial PMF has a parameter vector $\phi = [\phi_1, \phi_2, \phi_3, \phi_4, \phi_5]$, which does not depend on γ_0 or γ_1 , so they can be ignored in the following equation:

$$P(n|\gamma_0, \gamma_1, \alpha, \beta) = P(n|\alpha, \beta) = P_{\text{MN}}(n|\phi)$$

Where P_{MN} denotes a multinomial PMF. $P(n|\alpha, \beta)$ can be read as the probability of the attempt counts based on the Beta-distributed distances.

To complete the Bayesian model, priors must be placed on the four parameters. The parameters domain is as follows: $\gamma_0 \in (-\infty, \infty)$, $\gamma_1 \in (-\infty, \infty)$, $\alpha \in (0, \infty)$, and $\beta \in (0, \infty)$. These are uniform, flat priors for γ_0 and γ_1 , assuming total ignorance and improper priors over α and β to assume independence and favoring no particular scale. The uninformative prior can be written as follows:

$$f_{\gamma_0, \gamma_1, \alpha, \beta}(\gamma_0, \gamma_1, \alpha, \beta) \propto \frac{1}{\alpha\beta}$$

An uninformative prior is chosen because we do not have nor want prior beliefs to dominate the inference. This, in turn, gives more weight to the observed data in shaping the posterior distribution.

Due to the joint posterior distribution over the four model parameters being nonlinear, Laplace's approximation must be applied to estimate the posterior mode. In order to make this optimization more stable and improve the accuracy, the model should be reparametrized using the logarithms of the strictly positive parameters, being α and β . Therefore we use Laplace's approximation which is based on the negative log-posterior, so Baye's rule is rewritten as:

$$\begin{aligned} \ln f_{\gamma_0, \gamma_1, \ln \alpha, \ln \beta | m, n}(\gamma_0, \gamma_1, \ln \alpha, \ln \beta | m, n) \\ \propto \ln P(m, n | \gamma_0, \gamma_1, \ln \alpha, \ln \beta) + \ln f_{\gamma_0, \gamma_1, \ln \alpha, \ln \beta}(\gamma_0, \gamma_1, \ln \alpha, \ln \beta) \end{aligned}$$

The log-likelihood is then broken down into the two terms:

$$\ln P(m, n | \gamma_0, \gamma_1, \ln \alpha, \ln \beta) = \sum_{k=1}^K \ln P_{\text{Bin}}(m_k | n_k, p_k) + \ln P_{\text{MN}}(n | \phi)$$

Because the prior is uninformative and reparametrized using $\ln \alpha$ and $\ln \beta$, it ends up proportional to 1, so the log-posterior is just proportional to the log-likelihood:

$$\ln f_{\gamma_0, \gamma_1, \ln \alpha, \ln \beta | m, n}(\gamma_0, \gamma_1, \ln \alpha, \ln \beta | m, n) \propto \sum_{k=1}^K \ln P_{\text{Bin}}(m_k | n_k, p_k) + \ln P_{\text{MN}}(n | \phi)$$

To evaluate each p_k , which involves an integral over a logistic function weighted by a Beta density, MATLAB's trapz function is utilized over the interval $[a_k, b_k]$ to compute the necessary numerical integrations.

In order to model the logistic regression of expected field goal distance, Bayesian inference was implemented in MATLAB using the fminunc function by maximizing the log-posterior creating the *Maximum A Posteriori* estimate, which identifies the parameter values that maximize the

posterior probability given the observed data. To associate each bin with a representative distance, I calculate the expected yardage using:

$$E[X|bin_k] = \frac{\int_{a_k}^{b_k} x f_X(x) dx}{\int_{a_k}^{b_k} f_X(x) dx}$$

This provides a more statistically accurate midpoint than simply taking the bin center. Using the expected yardage as the predictor and observed success rate in each bin as the response, the logistic regression model was fit with the `fitglm` function in MATLAB. The number of attempts n_k in each bin was used as a weight to reflect confidence in each bin's observed success rate.

The mean distance (average field goal attempt), mode (most common attempt) and standard deviation were calculated using the following equations respectively:

$$\begin{aligned} & \frac{\alpha}{\alpha + \beta} \\ & \frac{\alpha - 1}{\alpha + \beta - 2} \\ & \sqrt{\frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}} \end{aligned}$$

Results:

Field goal attempts and makes were categorized into five distance bins, normalized to the [0, 1] interval and corresponding to the following field distances in yards (2024 NFL Standings & Team Stats).

Table 1: NFL Database Distance Bin Data and Success Rate (2024 NFL Standings & Team Stats).

Distance Bin (yds)	Makes (FGM)	Attempts (FGA)	Success Rate
0-19.5	2	2	100.00%
19.5-29.5	229	236	97.03%
29.5-39.5	280	297	94.28%
39.5-49.5	231	301	76.74%
49.5-100	195	279	69.89%

From Table 1, it is shown that as the distance increases the success rate of kicks decreases. Furthermore, a significantly minimal number of kicks were attempted at lowest bin of 0-19.5 yards.

Table 2: Probability of Taking a Kick in Each Bin

Probability of taking a kick in each bin:
Bin 1 (0.0-19.5 yds): 0.0263
Bin 2 (19.5-29.5 yds): 0.1477
Bin 3 (29.5-39.5 yds): 0.2951
Bin 4 (39.5-49.5 yds): 0.2988
Bin 5 (49.5-100.0 yds): 0.2322

Using a Beta distribution to model attempt distances, the inferred probability of a kick falling into each bin is shown in Table 2. At the range of 0-19.5 yards, the probability of taking a kick is the lowest at around 2.6%, while the probability of taking a kick is the highest at the range of 29.5-39.5 yards.

Table 3: MAP Estimates of Beta Distribution for Field Goal Attempt Distances

=== Beta Distribution Parameters ===
Alpha (α) = 6.8163
Beta (β) = 9.8827

The alpha parameter increases the amount of attempts at shorter distances and the beta parameter increases the drop-off for longer-ranged kicks. From Table 3, it is apparent that the right tail is thinner than the left tail, meaning that beta is larger than the alpha value. This means that there are fewer very long kicks (>50 yards) than very short kicks (<20).

Table 4: Mean and Mode Distance with Standard Deviation

=== Distribution Statistics ===
Mean distance: 40.8 yards
Mode distance: 39.6 yards
Standard deviation: 11.7 yards

Using the alpha and beta parameters, the mean and mode distances as well as the standard deviation based on the Beta distribution were calculated in Table 4. Therefore, the average field goal attempt is around 40.8 yards, the most likely attempt distance is around 39.6 yards, and the spread of kick distances around the mean is 11.7 yards. This means that most field goal attempts fall within roughly 29-53 yards.

Table 5: Expected Yardage per Bin

```

=== Expected Yardage per Bin ===
Bin 1 (0.0-19.5 yds): 16.4 yards
Bin 2 (19.5-29.5 yds): 25.5 yards
Bin 3 (29.5-39.5 yds): 34.8 yards
Bin 4 (39.5-49.5 yds): 44.3 yards
Bin 5 (49.5-100.0 yds): 56.6 yards

```

Table 5 states the average field goal attempt distance within each bin computed using the Beta distribution. These values account for the shape of the distribution within each bin, not just the midpoint. These values were used to calculate the X values for the later logistic regression, mapping distance to success rate.

Table 6: Logistic Regression Coefficients Table (γ_0 and γ_1)

Logistic regression coefficients using expected yardage:

	Estimate	SE	tStat	pValue
(Intercept)	5.1404	0.40784	12.604	2.0043e-36
Distance	-0.078986	0.0084759	-9.3188	1.1762e-20

The intercept coefficient is γ_0 and the distance coefficient is γ_1 . γ_0 is the baseline value of the logistic function when distance is zero while γ_1 is the slope that controls how quickly success probability drops as distance increases. Because γ_1 is negative it shows that the chance of making a field goal decreases with distance (Table 6). The t-Stats measure how many standard errors the estimate is away from 0 with both values being large in magnitude. The p-Values for both are extremely small (close to zero) meaning that their effects are both statistically significant.

Table 7: Success Rate of Field Goal Attempt Given Distances

Distance (yds)	SuccessRate (%)
0	99.4
20	97.2
30	94.1
40	87.9
45	83
50	76.7
60	59.9

Using the p_k equation, the success rate of field goal attempts given distances was modeled using the calculated γ_0 and γ_1 values (Table 7). As distance increased success rate decreases.

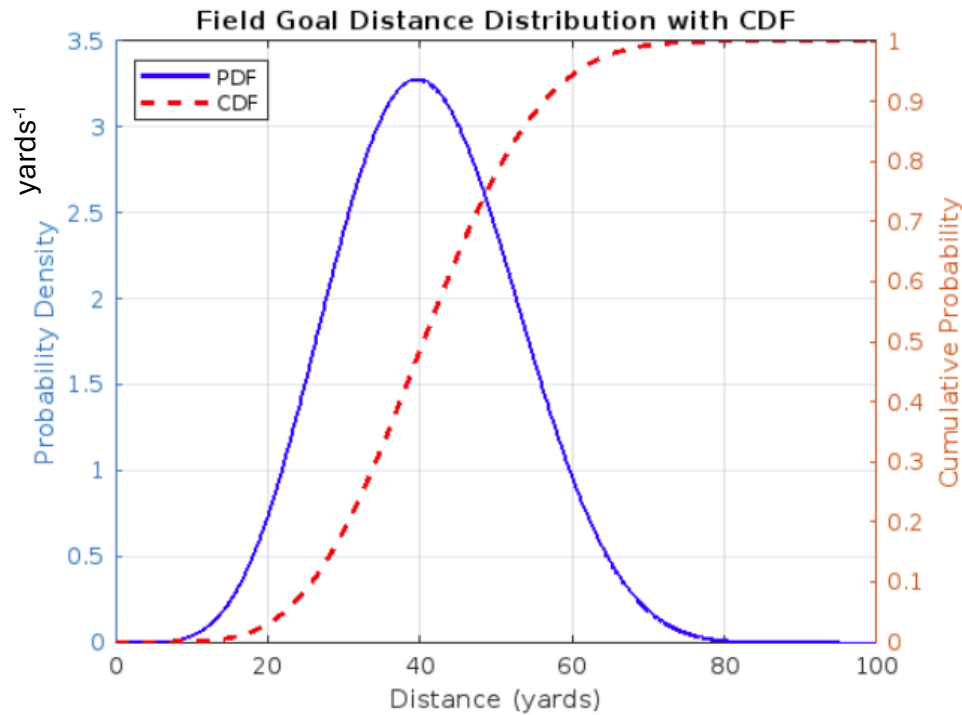


Figure 1: Probability Distribution of NFL 2024 Field Goal Attempt Distances

The blue curve shows the PDF, representing the relative likelihood of a kick being attempted at a given distance (Figure 1). The peak of the curve is around 39.6 yards, meaning kicks around this distance are most common. The distribution is slightly skewed to the right, which makes sense because very long field goals are rare but do occur. The red dashed curve is the CDF, showing the cumulative probability up to each distance. The CDF starts near 0 and approaches 1 as distance increase.

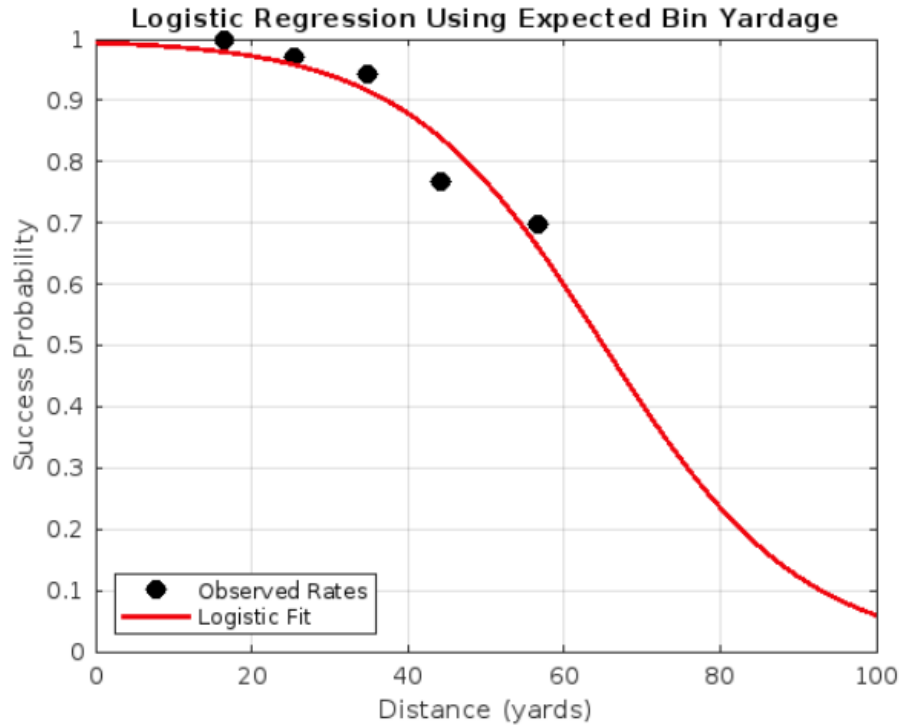


Figure 2: Logistic Regression Curve Using Expected Bin Yardage: Success Rate vs. Distance

This plot (Figure 2) shows the relationship between field goal success rate and kick distance modeled using logistic regression based on the equation:

$$p(x|\gamma_0, \gamma_1) = \frac{1}{1 + e^{-(\gamma_0 + \gamma_1 x)}}$$

Each black dot represents the empirical success rate in each of the five distance bins calculated from:

$$\text{Success Rate} = \frac{\text{Field Goals Made}}{\text{Field Goals Attempted}}$$

The x-values of these dots are the expected average distance within each bin, computed using the Beta distribution. The red line represents the predicted success probability from the logistic regression. It models how success probability decreases with increasing distance. The curve starts high near 1 for short distances, indicating high accuracy and gradually declines as distance increases, reflecting how longer kicks are harder to make.

Discussion:

This analysis used Bayesian modeling and logistic regression to better understand field goal accuracy in the NFL. It was found that field goals are attempted most often between 30 and 50 yards, with the highest probability of attempts occurring in the 39.5-49.5 yard bin (29.88%), followed closely by the 29.5-39.5 yard bin (29.51%) as found in Table 5. This suggests that coaches are most likely to attempt field goals in the intermediate range where success rates are still relatively high, but the reward in points is greater than for short range kicks. Together these bins account for nearly 60% of all field goal attempts. Interestingly, the probability of attempting a field goal in these two bins is nearly identical despite there being a 8.63% drop in success rate (of the expected means) according to the logistic regression model (Appendix A). Examining the extremes of the bins reveal an even higher drop of 22.3% (Table 7). This behavior may reflect game-time constraints or coaching tendencies to “settle” for longer kicks rather than risk turning the ball over on downs leaving their team in poor field position.

The Beta distribution fit to the attempt distances had a mean of 40.8 yards, a mode of 39.6 yards, and a standard deviation of 11.7 yards, confirming that most field goals are centered around the 40 yard mark with a moderate spread (Table 4). The logistic regression model reinforced that the probability of making a field goal declined steadily with distance with a negative slope of -0.079, meaning each additional yard significantly reduces the odds of success (Table 6). The regression intercept (5.14) corresponds to a high baseline success probability at short distances.

This analysis confirms what was already suspected: field goals become significantly more difficult the farther out they are attempted, and teams tend to avoid extremely short or long kicks unless necessary. Furthermore, it reveals how heavily penalties affect kicks as a 5 yard penalty at 45 yards causes an 83% to 76.7% success drop: a whopping 6.3% (Table 7).

Implications:

This model could be used in real time by coaching staffs or analysts to calculate the expected success rate of a field goal attempt based on its distance and to compare that with alternative options like going for it on 4th down. This could improve tactical decisions and potentially lead to better scoring outcomes. The results support using a data-driven approach in order to optimize decision making on whether to attempt a field goal, go for it on fourth down, or punt the ball away.

Limitations:

There are several limitations to this analysis including bin-based modeling, not accounting for weather conditions, dome vs outdoor conditions, or kicker skill. Because of how the data was collected and provided by the NFL, the analysis was forced to use bin-based modeling, assuming

uniform behavior within each yardage bin. If discrete data was provided for kicks at each specific yardage, a more accurate analysis could be provided. The logistic regression model only includes distance as a predictor of field goal success while in reality many other factors such as kicker skill, weather conditions, and field conditions all attribute to variance in success rate. By attributing all variance in success rate to exclusively distance, the model is potentially oversimplifying the decision-making process and underestimating uncertainty in predictions. Furthermore, game context is omitted as the model does not account for when and why a kick is attempted. A coach might attempt a low-probability kick at the end of the half or opt out of a high-probability kick in favor of a touchdown when trailing late in the game. Finally, including a bin as huge as 50-100 yards assumes they are behaviorally and statistically similar when they are not. A field goal past 70 yards has never been attempted in a game so grouping these bins together leads to misleading results.

Future Analysis

Potential future analyses that would extend the work done on this project would be to observe this data at discrete points instead of continuous bins. Furthermore, analysis of whether a team attempts a field goal or goes for it on 4th down across the NFL at certain yardages would further this analysis of seeing in what conditions teams should attempt field goals. Future analysis could involve compiling a dataset of all fourth down plays within field goal range 0-60 yard or continue with the 0-100 yard distribution and then classify each decision, field goal attempt, punt, or go for it. Estimation of expected points or win probabilities based on the chosen action in order to evaluate whether or not teams are making optimal choices given the data. This could ultimately support more data-driven coaching decisions and improve team strategy.

References

2024 NFL Standings & Team Stats. Pro. (n.d.). <https://www.pro-football-reference.com/years/2024/#kicking>

Dr Rego's Notes and Suggestions

Appendix A: Computed Success Probabilities and Percentage Drop

$$p(34.8) = \frac{1}{1 + e^{-(5.1404 - 0.078986 * 34.8)}} \approx 0.916$$

$$p(44.3) = \frac{1}{1 + e^{-(5.1404 - 0.078986 * 44.2)}} \approx 0.837$$

$$\% \text{ Drop} = \frac{.916 - .837}{.916} * 100 \approx 8.63\%$$

Appendix B: MATLAB Code (Proof It Worked At Least in MATLAB Grader)

```
1 % Data setup
2 interval_boundaries = [0, 19.5, 29.5, 39.5, 49.5, 100]; % In yards
3 a = interval_boundaries(1:end-1)/100; % Lower bounds (normalized)
4 b = interval_boundaries(2:end)/100; % Upper bounds (normalized)
5 n = [2, 236, 297, 301, 279]; % FGA data
6 m = [2, 229, 280, 231, 195]; % FGM data
7
8 % Run the Bayesian estimation (from previous code)
9 theta_0 = [logit(mean(m./n)); -0.1; log(2); log(2)];
10 options = optimoptions('fminunc', 'Display', 'iter', 'Algorithm', 'quasi-newton');
11 [theta_MAP, ~, ~, ~, ~, negH] = fminunc(@(theta) neg_log_post(theta, a, b, n, m), theta_0, options);
12 alpha_est = exp(theta_MAP(3));
13 beta_est = exp(theta_MAP(4));
14
15 %% 1. Probability of taking a kick in each bin
16 bin_probs = betacdf(b, alpha_est, beta_est) - betacdf(a, alpha_est, beta_est);
17 fprintf('\nProbability of taking a kick in each bin:\n');
18 for k = 1:5
19     fprintf('Bin %d (%3.1f-%3.1f yds): %.4f\n', k, a(k)*100, b(k)*100, bin_probs(k));
20 end
21
22 %% 2. PDF and CDF plot with statistics
23 x = linspace(0, 1, 1000);
24 pdf_values = betapdf(x, alpha_est, beta_est);
25 cdf_values = betacdf(x, alpha_est, beta_est);
26
27 % Calculate statistics
28 mean_dist = alpha_est/(alpha_est + beta_est) * 100; % Convert to yards
29 mode_dist = max(0, (alpha_est-1)/(alpha_est+beta_est-2)) * 100; % Mode formula for Beta
30 std_dist = sqrt(alpha_est*beta_est/((alpha_est+beta_est)^2*(alpha_est+beta_est+1))) * 100;
31
32 figure;
33 yyaxis left;
34 plot(x*100, pdf_values, 'b-', 'LineWidth', 2);
35 ylabel('Probability Density');
36 yyaxis right;
37 plot(x*100, cdf_values, 'r--', 'LineWidth', 2);
38 ylabel('Cumulative Probability');
39 xlabel('Distance (yards)');
```

```

40 title('Field Goal Distance Distribution with CDF');
41 grid on;
42 legend('PDF', 'CDF', 'Location', 'northwest');
43
44 % Add statistics to plot
45 text(10, 0.8*max(pdf_values), sprintf('Mean: %.1f yds\nMode: %.1f yds\nStd Dev: %.1f yds', ...
46     mean_dist, mode_dist, std_dist), 'BackgroundColor', 'white');
47
48 %% 3. Logistic regression using expected yardage in each bin
49 % Calculate expected yardage for each bin
50 bin_centers = (a + b)/2 * 100; % Midpoints in yards
51 observed_rates = m./n;
52
53 % Alternative: Calculate expected value within each bin
54 expected_yardage = zeros(1,5);
55 for k = 1:5
56     fun = @(x) x .* betapdf(x, alpha_est, beta_est);
57     integral_val = integral(fun, a(k), b(k));
58     expected_yardage(k) = integral_val/bin_probs(k) * 100;
59 end
60
61 % Fit logistic regression to expected yardage
62 % (Using MATLAB's fitglm for more robust fitting)
63 tbl = table(expected_yardage, observed_rates, 'VariableNames', {'Distance', 'SuccessRate'});
64 mdl = fitglm(tbl, 'SuccessRate ~ Distance', 'Distribution', 'binomial', 'Weights', n);
65
66 % Plot logistic regression
67 figure;
68 plot(mdl.Variables.Distance, mdl.Variables.SuccessRate, 'ko', 'MarkerSize', 8, ...
69     'MarkerFaceColor', 'k');
70 hold on;
71 x_range = linspace(0, 100, 100);
72 y_pred = predict(mdl, table(x_range, 'VariableNames', {'Distance'}));
73 plot(x_range, y_pred, 'r-', 'Linewidth', 2);
74 xlabel('Distance (yards)');
75 ylabel('Success Probability');
76 title('Logistic Regression Using Expected Bin Yardage');
77 legend('Observed Rates', 'Logistic Fit', 'Location', 'southwest');
78 grid on;

```

```
80 % Display regression coefficients
81 fprintf('\nLogistic regression coefficients using expected yardage:\n');
82 disp mdl.Coefficients;
83
84 %% Helper functions
85 function neg_log_post_pdf_val = neg_log_post(theta, a, b, n, m)
86     % Unpack and transform parameters
87     gamma_0 = theta(1);
88     gamma_1 = theta(2);
89     ln_alpha = theta(3);
90     ln_beta = theta(4);
91
92     alpha = exp(ln_alpha);
93     beta = exp(ln_beta);
94
95     % Fine grid for integration
96     dx = 1e-3;
97     xgrid = 0:dx:1;
98
99     % Precompute beta PDF and logistic function
100     f_X = betapdf(xgrid, alpha, beta);
101     p = 1./(1 + exp(-(gamma_0 + gamma_1*xgrid)));
102
103     % Number of intervals
104     K = numel(a);
105
106     % Proportion of X values in each interval
107     phi = betacdf(b, alpha, beta) - betacdf(a, alpha, beta);
108
109     % Success probability within each interval
110     p_k = nan(1, K);
111     for k=1:K
112         idx = xgrid >= a(k) & xgrid <= b(k);
113         p_k(k) = trapz(xgrid(idx), p(idx).*f_X(idx))/phi(k);
114     end
115
116     % Negative log-posterior
117     neg_log_post_pdf_val = -sum(log(binopdf(m, n, p_k))) - log(mnpdf(n, phi));
118 end
```

```
120 function y = logit(p)
121     % Logit function with protection for extreme values
122     p(p == 0) = 0.001;
123     p(p == 1) = 0.999;
124     y = log(p./(1-p));
125 end
126 %% Display key parameters and statistics
127 fprintf('\n=== Beta Distribution Parameters ===\n');
128 fprintf('Alpha ( $\alpha$ ) = %.4f\n', alpha_est);
129 fprintf('Beta ( $\beta$ ) = %.4f\n', beta_est);
130
131 fprintf('\n=== Distribution Statistics ===\n');
132 mean_dist = alpha_est/(alpha_est + beta_est) * 100;
133 fprintf('Mean distance: %.1f yards\n', mean_dist);
134
135 if alpha_est > 1 && beta_est > 1
136     mode_dist = (alpha_est-1)/(alpha_est+beta_est-2) * 100;
137 else
138     mode_dist = nan; % Mode doesn't exist in this case
139 end
140 fprintf('Mode distance: %.1f yards\n', mode_dist);
141
142 std_dist = sqrt(alpha_est*beta_est/((alpha_est+beta_est)^2*(alpha_est+beta_est+1))) * 100;
143 fprintf('Standard deviation: %.1f yards\n', std_dist);
144
145 fprintf('\n=== Expected Yardage per Bin ===\n');
146 for k = 1:5
147     fun = @(x) x .* betapdf(x, alpha_est, beta_est);
148     integral_val = integral(fun, a(k), b(k));
149     expected_yardage_bin = integral_val/bin_probs(k) * 100;
150     fprintf('Bin %d (%3.1f-%3.1f yds): %.1f yards\n', ...
151         k, a(k)*100, b(k)*100, expected_yardage_bin);
152 end
153 %% 4. Strategic Analysis
154
155 fprintf('\n=== Strategic Implications ===\n');
156
157 % Calculate success probabilities at key distances
158 key_distances = [0, 20, 30, 40, 45, 50, 60];
159 success_probs = 1./(1 + exp(-(5.1404 - 0.078986 * key_distances)));
160
161 fprintf('\nSuccess Probability at Key Distances:\n');
162 for i = 1:length(key_distances)
163     fprintf('%d yards: %.1f%% success\n', key_distances(i), success_probs(i)*100);
164 end
165
166 % Calculate probability drop for 5-yard penalty at long range
167 long_range = 45;
168 prob_45yd = 1/(1 + exp(-(5.1404 - 0.078986 * 45)));
169 prob_50yd = 1/(1 + exp(-(5.1404 - 0.078986 * 50)));
170 penalty_drop = (prob_45yd - prob_50yd)*100;
171
172 fprintf('\nA 5-yard penalty from 45 to 50 yards reduces success probability by %.1f%%\n', penalty_drop);
173
```

First-order

Iteration	Func-count	f(x)	Step-size	optimality
0	5	414.72		407
Objective function returned Inf; trying a new point...				
1	16	361.861	0.000538692	238
2	26	176.945	10	395
3	36	119.373	0.5	52.7
4	41	117.897	1	62.9
5	46	116.773	1	22
6	51	116.539	1	20.1
7	56	115.587	1	24.1
8	61	114.888	1	20.5
9	66	114.682	1	14.2
10	71	114.52	1	14.7
11	76	114.126	1	30.8
12	81	113.14	1	65.8
13	86	110.617	1	121
14	91	104.586	1	195
15	96	92.2711	1	259
16	101	77.2462	1	219
17	106	70.8341	1	90.9
18	111	69.7324	1	20.3
19	116	69.5605	1	1.62

First-order

Iteration	Func-count	f(x)	Step-size	optimality
20	121	69.5317	1	1.72
21	126	69.5293	1	0.496
22	131	69.5292	1	0.0173
23	136	69.5292	1	0.00422
24	141	69.5292	1	0.000803
25	146	69.5292	1	8.45e-05

Computing finite-difference Hessian using objective function.

Local minimum found.

Optimization completed because the size of the gradient is less than the value of the optimality tolerance.

<stopping criteria details>

Probability of taking a kick in each bin:

Bin 1 (0.0-19.5 yds): 0.0263

Bin 2 (19.5-29.5 yds): 0.1477

Bin 3 (29.5-39.5 yds): 0.2951

Bin 4 (39.5-49.5 yds): 0.2988

Bin 5 (49.5-100.0 yds): 0.2322

Logistic regression coefficients using expected yardage:

	Estimate	SE	tStat	pValue
	_____	_____	_____	_____
(Intercept)	5.1404	0.40784	12.604	2.0043e-36
Distance	-0.078986	0.0084759	-9.3188	1.1762e-20

=== Beta Distribution Parameters ===

Alpha (α) = 6.8163

Beta (β) = 9.8827

=== Distribution Statistics ===

Mean distance: 40.8 yards

Mode distance: 39.6 yards

Standard deviation: 11.7 yards

=== Expected Yardage per Bin ===

Bin 1 (0.0-19.5 yds): 16.4 yards
Bin 2 (19.5-29.5 yds): 25.5 yards
Bin 3 (29.5-39.5 yds): 34.8 yards
Bin 4 (39.5-49.5 yds): 44.3 yards
Bin 5 (49.5-100.0 yds): 56.6 yards

=== Strategic Implications ===

Success Probability at Key Distances:

0 yards: 99.4% success
20 yards: 97.2% success
30 yards: 94.1% success
40 yards: 87.9% success
45 yards: 83.0% success
50 yards: 76.7% success
60 yards: 59.9% success

A 5-yard penalty from 45 to 50 yards reduces success probability by 6.3%

